🩺 Document 3: Specialist Clinical Data & EMR Specification

Product Name: ClinikFlow (India Specialist Edition)
Document Version: 1.0.0
Focus: Touch-First Clinical Encounter Data Capture for Solo Specialists
Target Form Factors: iPad, Android Tablet, Smartphone (iOS & Android)

1. Design Philosophy: The 60-Second Consultation Encounter

In an Indian private OPD, doctors see between 30 and 60 patients in a single evening. If an EMR system requires typing with a physical keyboard or clicking through multiple popup dialogs, **doctors will abandon it immediately**.

ClinikFlow EMR Core Tenets:

1. **Zero Mandatory Typing:** 95% of routine clinical entries are captured via tap-chips, sliders, segmented pickers, and numeric steppers.
2. **Thumb-Friendly Ergonomics:** Main action buttons (`[Next]` , `[Hold]` , `[Prescribe]`) are placed within the thumb zone of a 10-inch tablet or 6.5-inch phone held in portrait or landscape.
3. **Specialty-Adaptive:** The UI changes its clinical data fields automatically based on the doctor's specialty (Pediatrics, Dermatology, Orthopedics, or Internal Medicine/Diabetology).

2. Universal Clinical Header & Vitals Pad

Every specialist consultation begins with a compact, collapsible vitals bar.

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ [← Back to Queue]      TOKEN #18: Sunita Devi (42F) • UHID: ST-2610-00412    │
├──────────────────────────────────────────────────────────────────────────────┤
│ VITALS QUICK-PAD:                                                            │
│ [ BP: 130 / 84 mmHg ]  [ Pulse: 78 bpm ]  [ Wt: 64.5 kg ]  [ Ht: 158 cm (BMI: 25.8) ]  │
│ [ Temp: 98.6 °F ]      [ SpO2: 99% ]      [ Blood Sugar: 142 mg/dL (RBS) ]   │
└──────────────────────────────────────────────────────────────────────────────┘
```

JSON Schema for Universal Vitals

```
json
{
  "bp_systolic": 130,
  "bp_diastolic": 84,
  "pulse_bpm": 78,
  "weight_kg": 64.5,
  "height_cm": 158.0,
  "bmi": 25.84,
  "temperature_f": 98.6,
  "spo2_percent": 99,
  "blood_sugar": {
    "value": 142,
    "type": "RBS" // Options: "FBS", "PPBS", "RBS", "HBA1C"
  }
}
```

3. Four Core Specialist Clinical Modules

Module A: Pediatrics (Child Care & Development)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ PEDIATRIC SPECIALIST DASHBOARD                                               │
├──────────────────────────────────────────────────────────────────────────────┤
│ • Age: 18 Months (Born: 14-Apr-2025)        • Head Circumference: 47.5 cm (50th %tile) │
│ • Weight-for-Age: 11.2 kg (60th %tile)      • Height-for-Age: 83.0 cm (55th %tile)     │
├──────────────────────────────────────────────────────────────────────────────┤
│ UPCOMING / DUE VACCINATIONS (Indian Academy of Pediatrics - IAP Schedule):   │
│ [X] BCG          [X] OPV 1,2,3      [X] Pentavalent 1,2,3    [X] Rotavirus 1,2,3  │
│ [X] PCV 1,2      [X] MMR 1 (9 Mo)   [!] DTP Booster 1 (DUE)  [!] MMR 2 (DUE)      │
│ Action: [ Tap to Mark Administered Today ]                                   │
├──────────────────────────────────────────────────────────────────────────────┤
│ DEVELOPMENTAL MILESTONES (18 Months):                                        │
│ [✓] Walks alone   [✓] Says 5-10 words   [✓] Uses spoon   [ ] Climbs stairs unsupported │
└──────────────────────────────────────────────────────────────────────────────┘
```

- **Data Schema:**

```
json
{
  "pediatrics": {
    "head_circumference_cm": 47.5,
    "growth_percentiles": { "weight": 60, "height": 55, "head_circ": 50 },
    "vaccines_given_today": ["DTP_BOOSTER_1", "MMR_2"],
    "milestones_achieved": ["walks_alone", "speaks_single_words", "uses_spoon"],
    "feeding_type": "WEANED_FAMILY_FOOD" // "BREASTFEEDING", "FORMULA", "WEANING"
  }
}
```

Module B: Dermatology (Skin, Hair & Nails)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ DERMATOLOGY SPECIALIST DASHBOARD                                             │
├──────────────────────────────────────────────────────────────────────────────┤
│ ANATOMIC REGION (Tap to Select):                                             │
│ [ Face / Cheeks ] [ Scalp ] [ Forearms ] [ Trunk / Back ] [ Lower Legs ] [ Intertriginous]│
├──────────────────────────────────────────────────────────────────────────────┤
│ LESION CHARACTERISTICS (Multi-Select Chips):                                 │
│ [ Erythematous ] [ Maculopapular ] [ Plaque ] [ Vesicular ] [ Pustular ] [ Scaling ]   │
│ [ Hyperpigmented ] [ Annular Border ] [ Lichenified ] [ Pruritus Score: 7/10 ]        │
├──────────────────────────────────────────────────────────────────────────────┤
│ CLINICAL PHOTOGRAPHY:                                                        │
│ [ 📷 Take Photo (Camera) ]   [ 🟨 Attached: IMG_001.jpg (Right Malar Rash) ]          │
└──────────────────────────────────────────────────────────────────────────────┘
```

- **Data Schema:**

```
json
{
  "dermatology": {
    "affected_regions": ["FACE_CHEEKS", "FOREARMS"],
    "lesion_morphology": ["ERYTHEMATOUS", "PLAQUE", "SCALING"],
    "pruritus_vas_score": 7,
    "clinical_photos": [
      { "url": "s3://clinik-uploads/photos/img_001.webp", "caption": "Right Malar
Rash", "timestamp": "2026-10-14T18:40:00Z" }
    ]
  }
}
```

Module C: Orthopedics (Joints, Spine & Trauma)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ ORTHOPEDIC SPECIALIST DASHBOARD                                              │
├──────────────────────────────────────────────────────────────────────────────┤
│ JOINT / SPINE LOCATOR:                                                       │
│ [ Right Knee ] [ Left Knee ] [ Lumbar Spine (L4-L5) ] [ Cervical Spine ] [ Right Shoulder
]│
├──────────────────────────────────────────────────────────────────────────────┤
│ PHYSICAL EXAM FINDINGS:                                                      │
│ Tenderness:   [ None ] [ Mild (+) ] [ Moderate (++) ] [ Severe (+++) ]       │
│ Swelling:     [ Present ] [ Absent ]       Joint Crepitus: [ Present ] [ Absent ]  │
│ Range of Motion: [ Full & Free ] [ Painful at Terminal Flexion ] [ Restricted < 90° ]  │
├──────────────────────────────────────────────────────────────────────────────┤
│ COMMON RADIOLOGY ORDERS (Tap to Order):                                      │
│ [✓] X-Ray Knee AP & Lateral Standing     [ ] MRI Lumbar Spine                │
│ [✓] Serum Uric Acid & Calcium             [ ] DEXA Bone Density Scan         │
└──────────────────────────────────────────────────────────────────────────────┘
```

Module D: Diabetology & Internal Medicine

```
┌──────────────────────────────────────────────────────────────────────────┐
│ DIABETOLOGY & CARDIOMETABOLIC DASHBOARD                                  │
├──────────────────────────────────────────────────────────────────────────┤
│ COMORBIDITIES (Tap to Toggle):                                           │
│ [✓] Type 2 Diabetes (Duration: 8 yrs)   [✓] Hypertension   [ ] CAD / Post-PTCA │
│ [✓] Dyslipidemia                        [ ] CKD Stage 1-3  [ ] Diabetic Neuropathy │
├──────────────────────────────────────────────────────────────────────────┤
│ GLYCEMIC TARGETS & TRENDS:                                               │
│ Last HbA1c: [ 7.8% (Aug 2026) ]   Today RBS: [ 164 mg/dL ]   Target HbA1c: [ < 7.0% ] │
│ Foot Examination: [ Monofilament 10g Intact ] [ Peripheral Pulses Palpable ] │
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```

4. The Rapid Indian Prescription (Rx) Engine

1. Autocomplete Search Database

Pre-populated with top 3,000 Indian pharmaceutical brands, strengths, and generics:

- E.g. Typing `Aug` shows:
  - `Augmentin 625 Duo` (Amoxicillin 500mg + Clavulanic Acid 125mg)
  - `Augmentin 375`
  - `Augmentin DDS Suspension` (Pediatric)

2. Tap-to-Prescribe Chip Grid

When a drug is selected, dosage fields are instantly filled using **single-tap chips**:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│ MEDICINE: Augmentin 625 Duo Tablet (Amoxyclav)                               │
├──────────────────────────────────────────────────────────────────────────────┤
│ FREQUENCY:                                                                   │
│ [ 1 - 0 - 1 ] (Morning-Night)   [ 1 - 0 - 0 ] (Morning)   [ 0 - 0 - 1 ] (Night)  │
│ [ 1 - 1 - 1 ] (TDS)             [ SOS ] (When Needed)     [ Custom ]         │
├──────────────────────────────────────────────────────────────────────────────┤
│ MEAL TIMING:                                                                 │
│ [ After Food (खाने के बाद) ]    [ Before Food (खाली पेट) ] [ With Food (खाने के साथ) ] │
├──────────────────────────────────────────────────────────────────────────────┤
│ DURATION:                                                                    │
│ [ 3 Days ]  [ 5 Days ]  [ 7 Days ]  [ 10 Days ]  [ 15 Days ]  [ 1 Month ]    │
├──────────────────────────────────────────────────────────────────────────────┤
│ [ + Add to Prescription ]                                                    │
└──────────────────────────────────────────────────────────────────────────────┘
```

3. Prescription JSON Output

```
json
[
  {
    "brand_name": "Augmentin 625 Duo",
    "composition": "Amoxicillin 500mg + Clavulanic Acid 125mg",
    "form": "TABLET",
    "dosage_pattern": "1-0-1",
    "timing": "AFTER_FOOD",
    "duration_days": 5,
    "total_quantity": 10,
    "instructions": "Take with plenty of water. Do not skip doses."
  },
  {
    "brand_name": "Pantocid 40",
    "composition": "Pantoprazole 40mg",
    "form": "TABLET",
    "dosage_pattern": "1-0-0",
    "timing": "BEFORE_FOOD",
    "duration_days": 5,
    "total_quantity": 5,
    "instructions": "Take 30 minutes before breakfast."
  }
]
```

5. Digital Prescription PDF Specification

When the doctor taps `[Complete Consultation & Send Rx]`, the application compiles an official, tamper-proof PDF:

1. **Header (Doctor & Clinic Letterhead):**
    - Clinic Name, Doctor Name, MBBS / MD / DM Degrees.
    - State Medical Council Registration Number (e.g. `MCI-2012-04-1829`).
    - Clinic Address, Contact Number, Emergency Helpline.
2. **Patient Demographics Bar:**
    - Token No, UHID, Patient Name, Age, Gender, Date & Time.
3. **Clinical Summary:**
    - Chief Complaints & Recorded Vitals.
    - Provisional Diagnosis.
4. **Prescription Table ($\mathrm{R_x}$ Rx):**
    - Numbered list of medicines with composition, dosage, timing in vernacular, and duration.
5. **Footer:**
    - Investigation Advice & Follow-up Date.
    - Digital Signature / QR code to verify authenticity.
    - Statutory Disclaimer: "*Valid for 6 months from issue date. Substitution permitted with generic equivalents.*"